

- 2 (a) In the sweet pea plant, *Lathyrus odoratus*, one gene codes for flower colour and one gene codes for pollen grain shape.

Flower colour is either purple or red. Pollen grain shape is either long or round.

The inheritance of these genes is an example of **autosomal linkage**.

- The allele **F** for purple flowers is dominant over the allele **f** for red flowers.
- The allele **G** for long pollen grains is dominant over allele **g** for round pollen grains.

Explain the meaning of the term *autosomal linkage*.

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.....[2]

- (b) A dihybrid cross was carried out between homozygous dominant and homozygous recessive sweet pea plant parents to produce the F1 generation.

The offspring from the F1 generation were crossed to produce the F2 generation.

- (i) Draw a genetic diagram to show a dihybrid cross between two offspring from the F1 generation.

Assume that these genes are closely linked and that there are **no** crossing over events.

- (ii) The actual results of the dihybrid cross are shown in Table 2.1.

**Table 2.1**

| phenotypes of F2 offspring          | number of individuals |
|-------------------------------------|-----------------------|
| purple flowers, long pollen grains  | 284                   |
| purple flowers, round pollen grains | 21                    |
| red flowers, long pollen grains     | 21                    |
| red flowers, round pollen grains    | 55                    |

State how the results support the fact that this is an example of autosomal linkage.

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.....  
.....[1]

- (c) (i) In a test cross, an individual of **known** genotype is crossed with an individual that has a dominant phenotype but unknown genotype.

State the genotype of the **known** individual in a test cross.

.....[1]

- (ii) A test cross was carried out with sweet pea plants known to be heterozygous for both flower colour and pollen grain shape.

The results of the test cross are shown in Table 2.2.

**Table 2.2**

| phenotypes of offspring of test cross | number of individuals |
|---------------------------------------|-----------------------|
| purple flowers, long pollen grains    | 215                   |
| purple flowers, round pollen grains   | 30                    |
| red flowers, long pollen grains       | 32                    |
| red flowers, round pollen grains      | 210                   |

The result of a test cross can be used to determine a crossover value (COV).  
A crossover value is the percentage of the total number of offspring showing recombination.

The crossover value (COV) can be calculated using the formula shown in Fig. 2.1.

$$\text{COV} = \frac{\text{number of recombinants}}{\text{total number of individuals}} \times 100$$

**Fig. 2.1**

Calculate the COV from the results shown in Table 2.2.

COV = .....% [1]

- (iii) Suggest what information about the relative distance between the linked genes can be gained from crossover values.

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.....[1]

[Total: 10]